13. Priority, nice, and resource limits (ulimit)

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Phase: Phase 4 — Processes and jobs
Estimated time: 1 day
Slug: `13-nice-priority`

Learning objectives

- Adjust niceness
- Inspect ulimit
- Explain why limits exist
- Connect to cgroups at a high level

Prerequisites

Complete prior numbered lessons. Prefer a real Linux VM/container for Linux-specific topics.

nice

```
nice -n 10 ./batch.sh
renice 5 -p <pid>
```

Higher nice value → lower priority (be nicer to others).

ulimit

```
ulimit -a
ulimit -n 4096   # open files (session)
```

Production systems set limits via systemd (`LimitNOFILE=`) or `/etc/security/limits.conf`.

cgroups

Modern Linux isolates CPU/memory via cgroups (containers use this heavily).

Common mistakes

- Raising nofile without diagnosing fd leaks
- Assuming nice makes a slow disk fast
- Ignoring OOM killer symptoms

Exercises

Work in `exercises/13-nice-priority/`. Run:

```
./test.sh
```

- **A (Drill):** make `./test.sh` pass
- **B (Apply):** note one real command you ran on a Linux host related to this lesson
- **C (Challenge):** rewrite your solution from memory the next day

Do not open `solutions/13-nice-priority/` until you have an honest failing attempt.

Self-check

1. Does higher nice mean more CPU?
2. Why limit open files?
3. What tech do containers use for limits?

Further reading

- https://man7.org/linux/man-pages/man1/nice.1.html